

Allometric calibration bias in tropical forest carbon

Architecture-informed measurement, LiDAR dependence and annual greenhouse-gas attribution

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Abstract

Allometric calibration can misrepresent tropical forest architecture and organ-level carbon, and a remote-sensing model can inherit that error from its training labels. This article examines that measurement-to-accounting chain using architecture-informed reconstruction and the published 723.97 PgC pantropical reassessment as its methodological basis and application. Organ volume, basic density and tissue-carbon measurements are distinguished from merchantable timber volume and from reduced-input biomass prediction. A conditional-expectation identity shows how label bias persists in an ideal remote-sensing predictor, while a population-transfer identity specifies how changing architectural composition can create aggregate error. Greater LiDAR coverage therefore does not, by itself, establish carbon-scale accuracy; direct laser-based material reconstruction is a different pathway that can reduce allometric dependence. The original reconstruction, tissue analyses and physical-volume comparison are presented alongside independent architectural evidence, with their spatial and biological domains preserved. Exact stock-to-flux relationships then propagate carbon corrections through disturbance selection, delayed release, regrowth and finite census increments. Deforestation and degradation include all stem sizes through abundance and affected fractions. A positive omitted net source raises reconstructed atmospheric accumulation unless other changes offset it; a retained total instead constrains source reattribution. The reported stock retains its mapped pantropical extent, Australian-tropics exclusion and nominal 2007–2008 epoch. The synthesis separates evidence for the measurement approach from the additional observations needed to quantify present-day emissions, without deriving a new global flux from a standing-stock multiplier alone.

Keywords: allometric calibration bias; tropical forest carbon; tree architecture; lidar; reference-label bias; biological volume; tissue carbon; deforestation and degradation; stock-to-flux transfer.

1 Introduction

The reliability of forest-carbon accounting begins with the biological quantity assigned to trees and stands, not with the spatial precision of the map used to distribute it. Allometry-based carbon labels may omit or misrepresent architecture and organ-level composition even when tree position, diameter or canopy height is accurately measured. When those labels calibrate a remote-sensing product, increased coverage can reproduce the same carbon-scale error over a larger area. A sensor measures properties of the forest; the conversion of those properties into biomass and carbon remains a separate inference.

This article examines the propagation of structural and compositional bias in allometry-based forest-carbon estimation into spatial products and annual greenhouse-gas accounts. Architecture-informed measurement and the published pantropical reassessment provide the methodological basis and application [1, 2]. Their central mechanism is that the actual material of stems, branches and other organs can differ from the material represented by a generalized prediction. Measuring that architecture is consequently a way to diagnose allometric bias, rather than an unrelated explanation for the resulting discrepancy.

The published estimate is 723.97 PgC in aboveground vegetation over its mapped pantropical domain, excluding the Australian tropics and assigned a nominal 2007–2008 map epoch [1, pp. 23–24]. It is not a world-forest total or a 2025 emissions quantity. This study uses that reported result and its measurement evidence without replacing it with an allometry-dependent comparison. Subsequent calibration may refine the estimated stock. The distinct task addressed here is to identify how measurement bias reaches a map and how a supported carbon correction reaches annual sources, removals and source attribution.

The contributions are a mechanism-based account of allometric transfer error, a precise distinction between label-trained remote sensing and material reconstruction, and exact stock-to-flux and census relationships. Detailed source-table diagnostics and conditional stand-volume comparisons are supplied in the technical supplement, where their reproducibility role is separated from the main measurement thesis. The analytical results distinguish improvement of a carbon reference from improvement in its spatial prediction. Extending a biased proxy and validating it against the same proxy cannot establish independent carbon accuracy.

Central inference. Architecture and organ allocation can cause error in allometric biomass predictions. LiDAR coverage alone does not remove a bias inherited from reference labels; independently checked three-dimensional material reconstruction can help replace those labels. The reported pantropical stock supplies the application, while disturbance, decomposition, regrowth and other uptake determine its annual atmospheric consequences. Published measurements, analytical demonstrations and unperformed empirical tests are distinguished throughout.

The evidence synthesis is purposive rather than a meta-analysis. The author developed the architecture-informed methodology underlying the cited preprint. Original measurements, source-reported method comparisons and analytical identities are identified separately; no synthetic stand is represented as a new field observation. Figures 1–3 connect the reported inventory, the forest-carbon interpretation and the two calibration pathways.

2 Allometric calibration bias as a measurement problem

2.1 Architecture as a mechanism of allometric calibration bias

Aboveground biomass allometry predicts dry mass from a selected set of tree attributes. A representative formulation is $\hat{B}_A = g_\theta(D, H, \rho)$, where diameter, height and density provide informative but incomplete descriptions of a tree. An architecture-informed estimate instead resolves the material in its organs. Two individuals sharing the predictors of a reduced-input relationship can differ in crown extent, branch hierarchy, buttress form, damage and allocation among organs. A population-average prediction need not recover either individual's mass exactly. Systematic differences between the calibration population and the forest being assessed can then turn unresolved individual variation into aggregate bias.

Underrepresented branch or stem material is a physical mechanism of allometric underestimation. A systematic discrepancy can therefore arise within the woody material already assigned to an aboveground-biomass target. The relevant comparison is between predicted and actual material in the same woody and nonwoody compartments. This distinction also separates target completeness from estimator accuracy: an equation may be intended to predict total aboveground biomass and still represent the distribution of that biomass imperfectly.

Direct measurements substantiate this mechanism in defined populations. Goodman et al. harvested and weighed 51 trees in southwestern Amazonia; branches contained $44\% \pm 2\%$ of aboveground biomass, and the tested pantropical equations including height but omitting crown dimensions underestimated the sample total by 11–14%. Incorporating crown information improved prediction [3]. These findings support an architectural source of allometric error. Their sample effect is neither a universal correction

factor nor a ceiling on discrepancies in other populations. They also show that adding relevant predictors can improve a statistical relationship; architecture-informed measurement can replace an inadequate calibration or help construct a better one.

2.2 Conditional representation and transfer between populations

Let X contain the recorded predictors and let Z contain additional architecture and composition. Write true biomass as a square-integrable function $B = \phi(X, Z)$ on the defined pool. For an ideal squared-error predictor calibrated in population P , $g_P(x) = \mathbb{E}_P[B \mid X = x]$. Even this ideal predictor has residual variation

$$\mathbb{E}_P[(B - g_P(X))^2] = \mathbb{E}_P[\text{Var}_P(B \mid X)]. \quad (1)$$

The conditional-mean property gives zero mean prediction error in P ; unresolved variation alone is therefore not a proof of population bias. Its role changes when the conditional distribution of architecture or composition changes. For a target population Q on overlapping predictor support,

$$g_P(x) - \mathbb{E}_Q[B \mid X = x] = \int \phi(x, z) \{dP(z \mid x) - dQ(z \mid x)\}. \quad (2)$$

The expression is exact when the same biological response function is used and the expectations exist. Finite-sample fitting, predictor measurement errors and a misspecified functional form add further errors. The formula identifies a mechanism: using the same diameter and height range does not establish that the unrecorded architecture is distributed in the same way. A downward error requires the conditional biomass in the target population to exceed the calibration expectation, rather than being assumed from heterogeneity alone.

A synthetic example illustrates the distinction. Suppose trees share the recorded predictors but have two architectural states with biomass 100 and 200 arbitrary units. Equal calibration frequencies give a prediction of 150. If the target frequencies are 0.2 and 0.8, the target mean is 180 and the transferred prediction is 30 units lower, a 16.7% shortfall. Increasing the number of trees assessed with the unchanged rule does not alter that expected discrepancy. These are analytical values, not measured tropical forest corrections; the executable example is provided in the numerical companion.

Homogeneous species, ages and management can narrow the conditional variation that a fitted model must represent, but homogeneity is not the criterion defining the validity of allometry. Chave et al. evaluated equations against 4,004 harvested trees from 58 tropical sites and reported broad predictive performance [4]. Applicability is therefore established by validated transfer across architecture and composition, rather than by a crop-versus-natural-forest classification. Crop and plantation relationships require calibration checks, while natural-forest relationships can be useful within independently supported domains. Treating either as an independently measured carbon scale without testing that transfer is the methodological error addressed here. What heterogeneous tropical forests particularly demand is explicit testing of architecture, composition and disturbance states that reduced predictors may fail to distinguish.

2.3 The working hypothesis and its observable consequences

The central hypothesis is that structural and compositional underrepresentation in the allometric calibration of some tropical forest populations produces consequential errors in carbon stocks and the fluxes derived from them. The published architecture-informed reassessment supplies the methodological basis and pantropical application [1, 2]. Its reported discrepancies motivate measurement at the level where the error is proposed to arise: organs, trees and vegetation populations, followed by the spatial and temporal operations used in accounting.

The hypothesis has observable consequences. At matched tree and plot supports, independent reconstructed material and tissue carbon should explain signed residuals of the allometric estimator, with

residual structure related to architectural or compositional attributes omitted from that estimator. Replacing biased reference labels should alter remote-sensing carbon predictions even when the sensor observations are unchanged. Disturbance-weighted residuals should then predict the correction to the source population more directly than the unweighted mean correction across intact forest. These predictions admit localized support, mixed signs or failure; their measurement does not require treating conventional inventory agreement as the definition of correctness.

3 Why LiDAR coverage does not by itself resolve carbon-scale bias

3.1 Sensor observations, calibration labels and independent reference

LiDAR supplies geometric information through laser ranging, return waveforms or point clouds. Biomass and carbon require additional material or empirical relationships. The crucial distinction is therefore the *inference pathway*, not the sensor name. A height-based biomass regression calibrated on allometric plot estimates remains dependent on those reference estimates. Detailed laser reconstruction of material volumes combined with independently measured density and carbon fractions can instead reduce that dependence. Both pathways have uncertainties, but they are not the same measurement procedure.

The documented GEDI Level-4A Version-2 pathway relates waveform relative-height metrics to field-plot estimates of aboveground biomass density, using regional and plant-functional-type models. Its guide describes geographic cross-validation and flags observations outside predictor or response support [5]. These controls address transfer and prediction errors; validation against reference labels alone cannot expose a systematic error common to the labels. This observation is not an estimate of the bias of a particular GEDI release. It identifies a component of reference uncertainty that must be established from the underlying field methods.

Conversely, Calders et al. reconstructed woody volume from terrestrial laser scanning and combined it with basic density in an Australian eucalypt forest. Against 65 harvested reference trees, the tested allometric totals were 29.85–36.57% low, whereas the TLS reconstruction was 9.68% high [6]. This demonstrates a route through direct geometric reconstruction rather than the same biomass allometry; it also demonstrates that laser-derived reconstruction is not error-free. Registration, occlusion, segmentation, branch geometry, leaf–wood separation and material-property sampling require their own checks. Thus increased sensing coverage cannot cure a biased calibration merely by being extensive, while appropriately used LiDAR can be part of the architectural solution.

Table 1: Three distinct measurement pathways. The same sensor can participate in different calibration chains; independence is an attribute of the chain, not of spatial resolution alone.

Pathway	Carbon information	Principal calibration question
Field predictors plus biomass allometry	Empirical mass prediction from recorded attributes	Does calibration represent the target architecture, composition and disturbance?
Remote structure plus allometric reference labels	Spatial prediction of the reference biomass/carbon scale	Does validation test true material, or repeat the label-generating estimator?
Organ reconstruction plus material analysis	Volume, basic density and tissue carbon, then spatial extension	Are the volumes and properties independently measured and spatially representative?

3.2 Calibration-label inheritance

Let C denote true carbon on a fixed pool and support, $Y = C + b + \varepsilon$ the calibration label, and L the remote-sensing predictors. Here b is the label’s signed systematic discrepancy and $\mathbb{E}_P[\varepsilon | L] = 0$. In the

population limit of squared-error prediction, define $m_Y(l) = \mathbb{E}_P[Y | L = l]$ and $m_C(l) = \mathbb{E}_P[C | L = l]$. Linearity of conditional expectation gives

$$m_Y(l) - m_C(l) = \mathbb{E}_P[b | L = l]. \quad (3)$$

This is an exact conditional-mean identity, not a claim of a new statistical theorem or of universal consistency for a particular learning algorithm. If a predictor accurately reproduces Y , its discrepancy from C can still equal the conditional label bias. An average-zero label error independent of L behaves differently: it contributes noise but no conditional-mean displacement in this ideal case. A target-population shift adds a second component,

$$m_Y(l) - \mathbb{E}_Q[C | L = l] = \mathbb{E}_P[b | L = l] + \{\mathbb{E}_P[C | L = l] - \mathbb{E}_Q[C | L = l]\}. \quad (4)$$

Separate reference-mass validation and transfer diagnostics are consequently required. Where predictors cannot distinguish structurally different trees or plots, accurate ranging cannot supply the omitted material information through a response label that already misrepresents it.

A complementary shared-error example makes the coverage distinction explicit. If $Y_i = C_i + b_0 + \varepsilon_i$, with a fixed shared offset b_0 and independent zero-mean errors of variance σ^2 , then the mean error has mean-squared value

$$\mathbb{E}[(\bar{Y}_n - \bar{C}_n)^2] = b_0^2 + \sigma^2/n. \quad (5)$$

More samples reduce the independent component, not the squared shared offset. Correlated errors require their full covariance instead of σ^2/n . Actual wall-to-wall pixels are not independent reference measurements, and a denser point cloud should not be equated with n independent carbon calibrations. The result separates spatial detail, statistical precision and absolute carbon accuracy without denying the value of better sensing.

3.3 Replacing the reference rather than merely refining the map

The constructive response is to calibrate the carbon scale from independently checked architecture and tissue properties, while using remote sensing to extend those labels through a documented support domain. Randomly splitting outputs from one label-generating model only tests prediction of that model. Complete site and vegetation holdouts, independent material references, and signed residuals against both allometric and architecture-based estimates provide stronger evidence. The same safeguard applies when an architectural model supplies training labels: geometry must be validated before label replication is interpreted as physical accuracy.

Calibration improvements may change the regional or pantropical point estimate without invalidating the measurement argument. The published 723.97 PgC result remains the reported application; it is neither recalculated from a LiDAR product here nor made the value that future calibration must reproduce. The methodological objective is the fidelity of the material-to-carbon chain and the consequences of replacing biased reference quantities. Figure 3 summarizes the two routes from field information to spatial products and annual accounts.

4 Architecture-informed measurement and the pantropical application

4.1 The architectural measurement chain and its physical tests

A dimensionally explicit tree-carbon representation is

$$C_j = \sum_o V_{jo} \rho_{jo}^b f_{jo}^C, \quad (6)$$

where non-overlapping compartments o have green material volume V , basic density ρ^b defined as oven-dry mass per green volume, and dry-mass carbon fraction f^C . With consistent units, the product is carbon mass. Hollow space, bark, leaves and branches must not be double counted; roots do not belong in an aboveground-only target. Moisture conventions must be matched rather than multiplying hydrated mass by a dry-mass carbon fraction.

Pantropical allometries can incorporate diameter, height and wood density [4], yet trees sharing those predictors can differ in crown architecture, branch allocation, hollow volume and tissue composition. A relationship based on the reduced predictors can describe their calibration-population average without recovering every individual's material distribution. Architecture-informed reconstruction resolves additional quantities, so its validity is not conditional on reproducing an allometric prediction. Tissue chemistry provides a separate measurement axis: a biomass correction and a carbon-fraction correction may have different signs [7]. The IPCC tropical aboveground default of 0.47 is an accounting parameter for specified applications, not a substitute for measured organ fractions [8].

NEB002 combines measured architecture, numerical integration or voxelization, material density and elemental carbon measurements; NEB001 links vegetation identity and environmental structure to spatial mapping [9]. Neural networks extend the resulting carbon estimates. The physical inputs therefore precede the statistical extension. Validation is assessed at each step: representation of measured material, assignment of density and chemistry, prediction beyond the reconstructed individuals, and integration across vegetation types. A model's ability to reproduce its training targets is relevant to the prediction step but does not summarize all the evidence supporting those targets.

The preprint reports 221 reconstructed individuals from 127 species (p. 6), and 2,232 tissue samples from 186 individuals processed with a LECO CHN-600 analyzer (pp. 16–17). Chemical quality control used reference compounds at ten-sample intervals. Its physical volume comparison used 10 individuals of *Aphelandra pulcherrima*, 1.54–2.9 m tall, with cutting, weighing and water displacement (p. 25 and Fig. 11). The comparison is summarized in Table 2; the measured material and the geometrical predictions are distinct observations [1].

Table 2: Reported physical-volume validation in the original preprint [1, p. 25]. The same 10 *Aphelandra pulcherrima* individuals provide the reference and reconstruction comparison. Mean volumes are source-reported; the final column gives the non-negative shortfall from the water-displacement reference, calculated from the displayed means and rounded to one decimal place. All three reconstructions fall below the reference, with the 1-mm voxel grid showing the largest shortfall. It is a difference of sample means, not an individual-error bound or a pantropical confidence interval.

Method	Mean volume (cm ³)	Shortfall from reference (%)
Water displacement (reference)	1,240.00	0.0
Numerical integration	1,217.76	1.8
Voxelization, 1-cm resolution	1,211.00	2.3
Voxelization, 1-mm resolution	1,189.92	4.0

This is physical validation of volume at the tested scale, alongside laboratory measurements of tissue carbon; it cannot be reduced to neural-network self-consistency. Its extension to different sizes and architectures is a distinct validation question. Numerical accuracy in integrating a specified mesh and accuracy of the mesh as a representation of a tree also remain separate quantities. Independent geometry, density and chemistry checks address these steps without requiring agreement with a conventional biomass equation.

The regional chain likewise precedes the pantropical total. The preprint reports network-based carbon estimates for more than 6,000 individuals in 27 vegetation types and a wider regional comparison across

54 vegetation types. Their reported average carbon densities were 177.03 MgC ha⁻¹ for the architecture-informed method and 61.41 for the allometric comparison. The pantropical calibration used regional carbon targets with the carbon map published by Baccini et al. in 2012 and the height map published by Simard et al. in 2011 [1, pp. 22–24]. The output has the preprint’s nominal 2007–2008 reference period, while predictor acquisition and calibration observations span different dates (Table 5). A conventional map used as a predictor is not thereby accepted as the true carbon target; shared predictors can nevertheless induce dependence between products. The complete chain is measured organs → reconstructed trees → vegetation-specific carbon estimates → spatial calibration, rather than a post hoc multiplication of an accepted global inventory.

4.2 Biological material is not merchantable timber

The physical target is the material in the defined biological compartments, irrespective of saleability. For disjoint woody components on the same live aboveground boundary,

$$V_W = V_{\text{merchantable}} + V_{\text{nonmerchantable}}. \quad (7)$$

Neither a minimum commercial top diameter nor timber quality removes existing carbon. Stem tops, branches, twigs, buttresses and noncommercial individuals must enter the target where present, with foliage or other nonwoody organs treated separately. Cavities contain no woody material; carbon remaining in degraded tissue must be represented with appropriate material properties and live/dead assignment rather than being excluded for lack of timber value. An outer crown envelope contains air and is not a material volume.

IPCC Box 4.2 explicitly distinguishes merchantable volume from total biomass and describes expansion of timber-volume records to additional components [8]. This is a definition of the measurement boundary, not validation of any default expansion factor. Direct biomass allometries may also target whole-tree aboveground dry mass, so it would be incorrect to equate every allometric output with saleable timber. The structural issue is whether the selected predictor or expansion actually represents the nonmerchantable material of the target vegetation.

For measured volume V and basic density ρ^b defined as oven-dry mass per corresponding green volume, $V\rho^b$ estimates dry mass; multiplying by the matched dry-mass carbon fraction gives carbon. Hydrated mass, green volume, dry volume and elemental-carbon fraction cannot be interchanged. Density and carbon fractions vary by organ and sample, and their products must be aggregated with the relevant material weights. A single assumed fraction is useful for an explicitly marked sensitivity calculation, not as a retroactive description of a method that measured tissue carbon directly.

Consequently, a hypothetical large unmeasured foliage remainder is not required to explain underestimated woody biomass. The disagreement can concern the woody material itself. Component completeness, prediction error within already included components, and spatial selection are distinct diagnostic questions. None should be used to replace the allometric-bias mechanism with an unsupported assumption that the conventional woody subtotal is exact.

4.3 Intermediate stems, abundance and forest degradation

Forest-carbon estimation and release accounting concern the full sampled size distribution. The architectural preprint compares organ biomass across diameters of 1–20 cm in Figure 6, extends its comparison to larger diameters in Figure 7, and presents carbon estimates for more than 6,000 individuals in Figure 8 [1, pp. 20–22, Figs. 6–8]. Its Table 1 contains, for example, aboveground-biomass estimates of 104.82 kg for *Inga* sp. 10 (15.24 cm DBH), 109.84 kg for *Pouteria* sp. 4 (11.58 cm DBH), and 115.74 kg for *Virola elongata* (14.63 cm DBH). These are biomass outputs, whereas Figure 8 uses kilograms of carbon. The role

of intermediate stems is therefore part of the original measurement domain, rather than an extrapolation from a large-tree validation sample.

A complementary stand-structure result appears in Vásquez and Arellano [10, Table 170, p. 942]. In the *Acalypha* sp.–*Guazuma ulmifolia* community, the 10–30 cm class contributes a reported 38.97% of aboveground biomass and 38.79% of carbon. Dividing its 2.88 tonnes of biomass by 30 stems per 0.05 ha gives 96 kg biomass per stem (Table 3). This ratio of displayed class aggregates illustrates an approximately 100 kg mean, not an assertion that every stem has that mass. Classes II and III together account for 62.71% of that community’s biomass. “Intermediate” here follows the local diameter classes; it is not a universal mass threshold.

Table 3: A published intermediate-stem example: the *Acalypha* sp.–*Guazuma ulmifolia* community in Córdoba [10, Table 170, p. 942]. Class II is 10–30 cm DBH in the table. Inputs and percentage shares are source-reported; means per stem are derived from the displayed aggregates. The chapter’s abstract gives a different lower-class interval, so the table’s explicit bin definition is used here.

Quantity	Value	Basis
Stems per 0.05 ha	30	Reported
Biomass (tonnes per 0.05 ha)	2.88	Reported
Carbon (tonnes C per 0.05 ha)	1.30	Reported
Share of community biomass (%)	38.97	Reported
Share of community carbon (%)	38.79	Reported
Mean biomass per stem (kg)	96.00	Derived
Mean carbon per stem (kgC)	43.33	Derived
Classes II + III biomass share (%)	62.71	Reported

The 2012 chapter estimated biomass using selected Chave et al. relationships and measured tissue-carbon fractions; the later architectural reconstruction is a distinct method. Its *Acalypha*–*Guazuma* community total is 7.39 Mg per 0.05 ha, or 147.8 Mg ha^{−1}, placing the abundance example in a comparatively low-biomass stand within that study [10, Table 170, p. 942]. It illustrates a size-class distribution, rather than supplying independent support for the class-5 density. The near-100 kg architectural records instead establish the size coverage of that method. Together with stands where larger classes dominate, the examples motivate vegetation-specific population weighting.

For disjoint classes h , let n_h be stem abundance and C_j the carbon of an individual. Then

$$C_h = \sum_{j \in h} C_j = n_h \bar{C}_h, \quad F_h(t) = \frac{44}{12} \sum_{\tau \leq t} \sum_{j \in \mathcal{D}_{h,\tau}} \sum_o C_{jo,\tau} d_{jo,\tau} k_{jo,\tau}(t - \tau).$$

Here $\mathcal{D}_{h,\tau}$ indexes affected stems, o indexes disjoint organs, $0 \leq d_{jo,\tau} \leq 1$ is the affected fraction, and k is the release rate as CO₂ in yr^{−1}. Thus stocks in kgC give release in kgCO₂ yr^{−1}. Repeated disturbances must use the remaining carbon and disjoint affected cohorts. Total forest release is $F = \sum_h F_h$: per-tree mass, abundance, exposure and release history all enter. A class containing lighter but numerous affected stems can supply more carbon or annual release than a sparse large-tree class. Partial crown and branch losses, collateral mortality, understory removal and repeated damage belong in degradation accounting even when land remains classified as forest. Regrowth and changes in future uptake are separate terms. These definitions are consistent with the cohort formulation in Section 6 and degradation-process distinctions in Lapola et al. [11].

5 Spatial, temporal and accounting boundaries

5.1 Stocks, transfers and gas-equivalent quantities

Aboveground biomass is dry biological mass; aboveground carbon is the carbon contained in that mass. Although $1 \text{ Pg} = 1 \text{ Gt}$, a stock in PgC cannot be inserted into a flow in GtC yr^{-1} . For carbon actually emitted as CO_2 , the conversion is

$$E_{\text{CO}_2} = (44/12)E_C. \quad (8)$$

It preserves, rather than creates, the time dimension. Movement from live wood to dead wood or harvested products is not necessarily an atmospheric emission in the same year. Carbon pools, transfers and eventual release therefore require explicit representation [8].

A multi-gas inventory instead applies a selected climate-equivalence convention,

$$T_{\text{GHG}} = \sum_g \text{GWP}_{100,g} E_g. \quad (9)$$

This is not a conserved carbon-mass balance. A CO_2 component can legitimately form the numerator of a CO_2e share because the equivalence factor for CO_2 is one; the denominator must still specify the other gases, time horizon and accounting boundary. Gas conversion does not make a net land-use category equivalent to gross tropical deforestation.

Throughout, a *gross component* means an explicitly positive emission or removal within the selected account. It does not mean that an anthropogenic inventory includes all ecosystem respiration or gross primary production. Those ecosystem fluxes cannot be added indiscriminately to an inventory of human-induced emissions.

5.2 The contemporary reference

Table 4: Reported reference quantities through 2025. Values are rounded source estimates or projections, not recalibrated forest-carbon results. LULUCF denotes land use, land-use change and forestry.

Quantity	Annual value		Boundary and status
EDGAR GHG excluding LULUCF	54.1 GtCO_2e		2025 inventory; AR5 GWP_{100}
EDGAR GHG including net LULUCF	Approximately 55.0 GtCO_2e		Same reported account including its land balance
EDGAR Forest Land	−5.5 GtCO_2		Living-biomass net removal; remaining forest and land converted to forest
EDGAR Deforestation	+3.7 GtCO_2		Net category, all regions; includes tropical forest-fire CO_2
EDGAR Fires	Approximately 2.2 GtCO_2e		Allocated fire account; tropical forest-fire CO_2 excluded to avoid overlap
EDGAR Organic Soils	About +1.1 GtCO_2		Reported CO_2 subtotal; the category also includes N_2O
EDGAR Other	See signed balance		Non-biomass forest pools and remaining land categories; not independently extracted here
EDGAR net LULUCF	About +0.9 GtCO_2e		Sum across the five categories above
GCB net land-use CO_2	About +4.1 GtCO_2		2025 projection, low confidence; different attribution convention

The land entries in Table 4 follow the *booklet*, its Section 3, Figure 6 and Annex 2, rather than a press-release paraphrase [12]. Forest Land already includes land converted to forest; afforestation and

regrowth must not be appended as a second removal subtotal. Tropical forest-fire CO₂ is assigned to Deforestation, whereas its non-CO₂ gases remain in Fires. Crop-burning CH₄ and N₂O are excluded from Fires where assigned to agriculture. The fire account does not distinguish managed from unmanaged land. These are inventory allocation rules, not a decomposition into all ecosystem gross fluxes.

The rounded entries yield the diagnostic remainder $0.9 - [3.7 - 5.5 + 2.2 + 1.1] \approx -0.6 \text{ GtCO}_2\text{e yr}^{-1}$. Because the organic-soil value quoted in the text is a CO₂ subtotal and the inputs are rounded, this is *not* an independently recovered numerical estimate of Other; it also absorbs omitted subcomponents and rounding. Figure 1 keeps the five categories distinct and labels this remainder accordingly. Land flows do not feed the fossil-CO₂ block. The non-LULUCF mixture is 73.8% fossil CO₂, 17.6% CH₄, 5.3% N₂O and 3.3% fluorinated gases; the power-and-heat category contributes nearly 30% of the non-LULUCF total [12].

The forest/deforestation estimates are a global, geographically explicit implementation of the IPCC Tier 1 Gain/Loss approach. They combine land-change information with default growth, biomass-expansion, wood-density, carbon-density and root-to-shoot parameters. Annex 2 describes forest-cover inputs through 2022 with subsequent estimates described by the booklet as interpolation, and a deforestation-driver product for 2001–2021; the label “2025 estimate” does not imply that every input was measured in 2025. Other additionally draws on country-reported data. This provenance is relevant to the same density-compatibility scrutiny applied to the historical benchmark [12].

The contrast between the approximately $+0.9 \text{ GtCO}_2\text{e yr}^{-1}$ EDGAR balance and $+4.1 \text{ GtCO}_2 \text{ yr}^{-1}$ GCB projection makes attribution conventions consequential to interpretation [12, 13]. Their nominal numerical separation of about 3.2 gigatonnes is not a matched-estimand discrepancy: gas coverage, managed-land attribution and estimation procedures differ. A direct harmonization study supplies the stronger quantitative evidence. For 2000–2020, Grassi et al. report $+4.8 \text{ GtCO}_2 \text{ yr}^{-1}$ from bookkeeping models and -1.9 from national inventories, a 6.7 gap. Including the model-estimated environmental-change sink in managed forests (-6.4) reduces the aggregate difference to $0.3 \text{ GtCO}_2 \text{ yr}^{-1}$ [14]. This demonstrates the importance of definitions in that comparison; it neither assigns the whole 2025 difference to conventions nor measures a structural biomass correction. A carbon account is determined jointly by physical measurement, pool/process boundaries and attribution rules. The paired-transfer analysis therefore specifies a physical account before comparing its alternative reporting conventions.

5.3 Pantropical coverage, worldwide extension and map dates

The reported total is a pantropical spatial integration, not an estimate for every forest worldwide. The source explicitly excludes the Australian tropics, and its classification includes forests together with other covered vegetation formations [1, pp. 23–24, Table 3 and Fig. 9]. Baccini et al. likewise map aboveground live woody vegetation across tropical America, Africa and Asia, excluding Australia; their coverage includes woody formations outside forest-only classes [15, p. 182 and Fig. 1]. A global height predictor does not extend the geographic support of the calibrated carbon output automatically.

Let P be the actual mapped pantropical domain and $W \supseteq P$ a harmonized worldwide aboveground-vegetation domain. For the same carbon pool and reference epoch t_0 , spatial additivity gives

$$C_W^{\text{AG}}(t_0) = C_P^{\text{AG}}(t_0) + C_{W \setminus P}^{\text{AG}}(t_0).$$

Retaining 723.97 PgC for $C_P^{\text{AG}}(t_0)$, a complete estimate on this wider support includes additional positive stock outside P , including the Australian tropics and temperate and boreal regions. Its increment is not quantified by the published pantropical total. Area-ratio scaling or transfer of one tropical correction factor to all other vegetation would introduce unsupported density and calibration assumptions. For a strictly forest-only target F , the appropriate partition is instead

$$C_F^{\text{AG}}(t_0) = C_{P \cap F}^{\text{AG}}(t_0) + C_{F \setminus P}^{\text{AG}}(t_0),$$

so the mapped forest contribution must first be distinguished from other included vegetation. Roots, dead organic matter and soil carbon are additional pools, not quantities already represented by the aboveground total.

The temporal reference is explicit rather than inferred from a bibliography year. Page 23 of the preprint assigns the calibrated map to 2007–2008 [1]. Baccini et al.’s carbon-density map is also described as representing 2007–2008, although the paper appeared in 2012 and its field calibration measurements span 2008–2010 [15, p. 182]. Simard et al.’s height predictor, published in 2011, uses GLAS laser-L3C observations from 20 May to 23 June 2005 and 2005 MODIS tree cover [16, Section 2.2]. The output’s nominal epoch therefore summarizes a multi-date calibration chain; it does not assert that every input, reconstruction or laboratory observation was collected during 2007–2008. The mapped quantity is standing carbon associated with that nominal reference period, not carbon accumulated during a two-year interval. Publication dates likewise do not update the mapped stock to 2012, 2016 or 2025.

Table 5: Spatial and temporal provenance of the pantropical stock estimate. Publication year, predictor acquisition and nominal output epoch have different meanings. The 2025 global emissions account in Table 4 is a separate annual quantity.

Source / publication	Spatial role	Reference period	Interpretation
Baccini et al. (2012)	Pantropical carbon-map predictor; Australia excluded	Map: 2007–2008; field calibration: 2008–2010	2012 dates the publication, not the mapped carbon.
Simard et al. (2011)	Global canopy-height predictor	GLAS L3C and tree cover: 2005	Predictor coverage and acquisition date do not redefine the carbon output.
Arellano-P. and Rangel-Ch. (2015/2016)	Calibrated pantropical carbon; Australian tropics excluded	Nominal map epoch: 2007–2008	Published 723.97 PgC result; preprint p. 23 specifies the epoch.

Geographic completion and temporal updating are separate operations. On a fixed, consistently tracked pool and mask, moving from t_0 to a later date requires the intervening gains, losses and pool transfers. Extending annual accounts worldwide instead sums the signed regional contributions: $N_W(t) = \sum_d [E_d(t) - U_d(t)]$ over disjoint regions and aligned attribution rules. Additional regions add positive stock and may add both emissions and removals; their net annual contribution can have either sign. Thus a larger worldwide standing stock does not, by itself, establish a larger 2025 net land flux or a proportional change in greenhouse-gas shares.

6 The stock-to-flux transfer identity

6.1 Matched support and fixed release operators

Consider spatial or cohort units $i = 1, \dots, n$ with positive baseline carbon C_i , local correction factors $r_i \geq 0$, and non-negative annual release weights w_i . The units, carbon pools and disturbance histories are held fixed. The weight has dimensions yr^{-1} when a stock is converted to an annual flow. Define

$$\mu_i = \frac{C_i}{\sum_j C_j}, \quad r_C = \sum_i \mu_i r_i, \quad r_F = \frac{\sum_i C_i r_i w_i}{\sum_i C_i w_i}, \quad (10)$$

with $\mathbb{E}_\mu[w] > 0$. Carbon-to- CO_2 conversion cancels in this ratio if applied consistently.

Proposition 6.1 (Stock-to-flux transfer). *Under these conditions,*

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w)}{\mathbb{E}_\mu[w]}. \quad (11)$$

Consequently $r_F = r_C$ if and only if the stock-weighted covariance is zero. Moreover,

$$\min_i r_i \leq r_F \leq \max_i r_i, \quad |r_F - r_C| \leq \sigma_\mu(r) \text{CV}_\mu(w). \quad (12)$$

Proof. Equation (10) gives $r_F = \mathbb{E}_\mu[rw]/\mathbb{E}_\mu[w]$. Substitution of $\mathbb{E}_\mu[rw] = \mathbb{E}_\mu[r]\mathbb{E}_\mu[w] + \text{Cov}_\mu(r, w)$ proves the identity. A positive denominator gives the equivalence. The first bound follows because r_F is a convex combination of r_i with weights proportional to $C_i w_i$. The second follows from Cauchy–Schwarz and $\text{CV}_\mu(w) = \sigma_\mu(w)/\mathbb{E}_\mu[w]$. $\square$

The result is the covariance decomposition of a weighted mean, algebraically identical to the selection term of the Price equation [17]. Its scientific role is to identify the missing empirical quantity: the association between where carbon is misestimated and where that carbon is released. Statistical independence is sufficient but not necessary; zero covariance under the specified stock weights is the exact condition. A constant r_i is another sufficient condition. Neither condition follows from a ratio of published global totals on different masks.

Table 6: Two-unit counterexample. Baseline stocks are (50, 50) and corrections (1, 3) in both rows, so $r_C = 2$. Only the annual release location changes. Units are synthetic.

Annual release weights	$\mathbb{E}_\mu[w]$	$\text{Cov}_\mu(r, w)$	r_C	r_F
(0.02, 0)	0.01	−0.01	2	1
(0, 0.02)	0.01	+0.01	2	3

6.2 Census changes, offsets and state-dependent calibration

For a stable scalar correction, two-census stock change obeys $\Delta C^* = r\Delta C$. A time-varying calibration instead gives the exact decomposition

$$C_1^* - C_0^* = r_0(C_1 - C_0) + C_1(r_1 - r_0). \quad (13)$$

The second term must be interpreted rather than automatically labelled instrument drift. It can reflect a changed calibration procedure, but it can also arise from a fixed size- or age-dependent calibration evaluated at different biological states. A fixed additive offset cancels for an unchanged set of units; recruitment, mortality or support changes can prevent that cancellation. A stable multiplicative error generally changes stock differences. A stand-wide scalar therefore requires justification for heterogeneous trees and changing composition.

This matters for repeated-census forest sinks. Plot-network analyses estimate biomass dynamics and net live-biomass accumulation using repeated tree observations and biomass relationships [18]. A shared, stable multiplicative mass correction changes these inferred gains and losses, not only disturbance emissions. However, a net increase in live aboveground biomass is not gross primary production or automatically net ecosystem–atmosphere uptake. Mortality transfers mass into other pools; decomposition, roots, soils, respiration and exports are needed to close the atmospheric boundary [8]. The paired formulation below applies only after those quantities have been aligned.

6.3 Disturbance cohorts and carbon fate

For area $A_{ia\tau}$ disturbed by process a in year τ , let h index disjoint tree-size, organ and fate compartments, with pre-disturbance carbon density $c_{iah\tau}$ and affected fraction $d_{iah\tau}$. Set $L_{iah\tau} = A_{ia\tau}c_{iah\tau}d_{iah\tau}$. With $k_{iah}^{\text{CO}_2}(t - \tau)$ the annual release rate (yr^{-1}) for CO_2 in year t ,

$$F_{\text{CO}_2}(t) = \frac{44}{12} 10^{-9} \sum_{i,a,h,\tau \leq t} L_{iah\tau} k_{iah}^{\text{CO}_2}(t - \tau). \quad (14)$$

Here L is in MgC and the output is $\text{GtCO}_2 \text{ yr}^{-1}$. For annual bins, $k\Delta t$ is the released fraction with $\Delta t = 1$ yr; cumulative fate fractions must not exceed the available cohort carbon. Other fate fractions represent retained live biomass, dead material, products, charcoal, exports or carbon released in other gases. Their mass bookkeeping must conserve carbon over the specified horizon. Regrowth is a separate uptake term and cannot be represented by making a gross release kernel negative.

The fixed-operator identity is exact when the correction changes carbon amounts but not areas, disturbance fractions or fate kernels. When those inputs also change, r_F must be recomputed from the revised products rather than attributed exclusively to r_C . An index can include cohort and pool, so the identity is compatible with delayed release; it does not remove the need to measure those delays.

6.4 Size-dependent release and the timing of a stock correction

The opposite-order sign used below is Chebyshev's sum inequality for similarly or oppositely ordered sequences, here written under stock weights. Its application concerns physical release from a defined cohort, rather than a rule for booking inventory stock losses. Combustion measurements in a rainforest clearing distinguish markedly different completeness among fuel-size components [19]. This supplies a physical candidate for the covariance in Proposition 6.1, not a universal monotone law: diameter, moisture, decay, burn history and exposure must be controlled. Where structural underestimation is concentrated in large wood and those compartments burn less completely, their carbon correction receives less weight in initial emissions. Extraction of boles into retained products can reinforce the initial delay, provided products and exports are tracked rather than lost from the boundary [8].

Proposition 6.2 (Opposite ordering and cumulative release). *On a fixed affected-carbon cohort with weights $\mu_i = C_i/C_\Sigma$, let $p_{it} \geq 0$ be the fraction of compartment i emitted as CO_2 in annual bin t , and $P_i(H) = \sum_{t \leq H} p_{it}$. If $(r_i - r_j)(p_{i0} - p_{j0}) \leq 0$ for every pair, then, for positive initial release, $r_{F,0} \leq r_C$. For a horizon with positive cumulative release,*

$$r_{F,\leq H} = r_C + \frac{\text{Cov}_\mu(r, P(H))}{\mathbb{E}_\mu[P(H)]}. \quad (15)$$

If $P_i(\infty) = 1$ for every compartment, $r_{F,\leq H} \rightarrow r_C$. The same conclusion holds for any common positive limiting fraction.

Proof. The covariance identity

$$\text{Cov}_\mu(r, p_0) = \frac{1}{2} \sum_{i,j} \mu_i \mu_j (r_i - r_j)(p_{i0} - p_{j0}) \quad (16)$$

has non-positive summands under the ordering assumption. Apply Proposition 6.1 to p_0 and to $P(H)$. A constant positive limiting release fraction has zero covariance. Strict initial inequality holds if some positive-weight pair has a strictly negative product. $\square$

The stock reference here is *carbon affected in the matched cohort*, not all standing pantropical carbon. The premise is a joint ordering to be tested, not a consequence of tree size alone; diameter-dependent drying or repeated burning can alter it. Complete oxidation as CO_2 within the tracked boundary is stronger than clearing a stand. Persistent charcoal, retained products, export outside that boundary and other gaseous carbon can leave unequal limiting fractions and nonzero cumulative covariance. A delayed correction need not reappear in every subsequent year. Under complete accounting, its cumulative allocation does recover the stock multiplier.

A synthetic two-compartment example uses stocks (50, 50), corrections (1, 3) and first-year fractions (0.8, 0.1). Then $r_C = 2$ but $r_{F,0} = 55/45 = 1.222 \dots$. If the remaining fractions (0.2, 0.9) are

subsequently oxidized, their combined later multiplier is $145/55 = 2.636 \dots$, and the lifetime multiplier is $200/100 = 2$. These synthetic transfers describe a physical release history. Applying them to EDGAR requires identifying whether the target is conversion-year stock-loss booking or atmospheric release: the inclusion of tropical fire CO_2 in Deforestation does not itself make a decay kernel part of the inventory operator. Section 6.5 distinguishes those accounting conventions.

6.5 Annual mixtures, disturbance trends and inventory conventions

A year- t release account combines cohorts of different ages. Assume fixed compartment shares $C_i > 0$ per unit disturbance, fixed corrections r_i , annual fate fractions $p_{i,s} \geq 0$, and a non-negative disturbance history $D(t-s)$. For annual time bins, let

$$W_i(t) = \sum_{s \geq 0} D(t-s)p_{i,s}, \quad r_{F,t} = \frac{\sum_i C_i r_i W_i(t)}{\sum_i C_i W_i(t)} = r_C + \frac{\text{Cov}_\mu(r, W(t))}{\mathbb{E}_\mu[W(t)]}. \quad (17)$$

Here $C_i D$ is affected carbon, and division by the one-year bin width gives an annual rate. The common stock weights presume stable compartment composition; changing forest types or disturbance selection require a cohort-specific extension.

Corollary 6.3 (Stationary mixtures and a two-lag trend criterion). *If D is constant over all contributing ages and $\sum_s p_{i,s} = 1$ for every compartment, then $r_{F,t} = r_C$. With two lags, $p_{i,1} = 1 - p_{i,0}$, and $D_0 = D(t)$, $D_1 = D(t-1)$,*

$$r_{F,t} - r_C = \frac{(D_0 - D_1) \text{Cov}_\mu(r, p_0)}{D_1 + (D_0 - D_1) \mathbb{E}_\mu[p_0]}. \quad (18)$$

The denominator is assumed positive. Oppositely ordered corrections and initial releases therefore suppress the annual multiplier during increasing disturbance, enhance it during decreasing disturbance, and cancel under stationarity.

Proof. Stationarity and complete ultimate release make each W_i equal to the same constant. For two lags, $W_i = D_1 + (D_0 - D_1)p_{i,0}$. Substitution into (17) proves the formula and sign. $\square$

For the synthetic cohort with $C = (50, 50)$, $r = (1, 3)$ and initial fractions $(0.8, 0.1)$, constant clearing gives annual baseline and corrected releases of 100 and 200. Doubling the latest cohort gives 145 and 255, hence $51/29 \approx 1.759$; halving it gives 77.5 and 172.5, hence $69/31 \approx 2.226$. The single-cohort first-year suppression concerns the multiplier relative to the stock correction, not disappearance of additional release: the corrected first-year amount is still larger than the baseline. Its expression in the mixed annual account depends on disturbance history. Longer kernels and changing composition require the complete convolution rather than the two-lag sign rule.

An inventory stock-loss convention and an atmospheric-release history are different operators. EDGAR describes a Tier 1 Gain/Loss implementation, but its Annex 2 does not supply a size-resolved decay kernel for the deforestation category [12]. Under a conversion-year convention that assigns biomass losses immediately, the relevant weights are affected fractions and converted area; decomposition timing does not modify that booked category. The release convolution instead matters when comparing the account with actual atmospheric transfers or with (31). A Tier label alone does not recover all implementation details.

Edge-associated biomass contrasts provide an empirical reason to measure the covariance rather than assume it vanishes. Chaplin-Kramer et al. report approximately 25% lower biomass within 500 m of tropical forest edges and estimate nearly 10% overstatement by Tier 1 carbon-stock methods that omit these effects [20]. These findings establish a spatial carbon contrast. If a uniform reference overstates edge stocks and independently observed clearing weights concentrate at those edges, the correction–disturbance covariance is negative. The biomass study alone does not measure that joint clearing distribution or establish a universal global sign.

7 Paired source–sink corrections

7.1 General reconciliation first; common-stock transfer as a special case

Let $F, U > 0$ be a specified source and positive removal magnitude, with corrected quantities supplied by appropriate, potentially different measurement operators. Define $r_F = F^*/F$, $r_U = U^*/U$ and the signed account $N = F - U + Z$.

Proposition 7.1 (Operator-independent net correction). *For any such corrected components,*

$$\boxed{\Delta N = (r_F - 1)F - (r_U - 1)U + \Delta Z.} \quad (19)$$

No common standing-stock basis or size-independent calibration is required.

Proof. Subtract $F - U + Z$ from $F^* - U^* + Z^*$ and substitute the two component ratios. $\square$

A more restrictive representation sets

$$F = \sum_i C_i w_i^+, \quad U = \sum_i C_i w_i^-, \quad w_i^\pm \geq 0, \quad (20)$$

with positive stocks, fixed weights and the same stock correction r_i multiplying both terms *within each unit*. Applying Proposition 6.1 separately gives

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w^+)}{\mathbb{E}_\mu[w^+]}, \quad r_U = r_C + \frac{\text{Cov}_\mu(r, w^-)}{\mathbb{E}_\mu[w^-]}. \quad (21)$$

Consequently, only in that shared-stock fixed-operator case,

$$\Delta N = (r_C - 1)(F - U) + C_\Sigma \text{Cov}_\mu(r, w^+ - w^-) + \Delta Z. \quad (22)$$

The proof is substitution in (19), using $F/\mathbb{E}_\mu[w^+] = U/\mathbb{E}_\mu[w^-] = C_\Sigma$. Spatially heterogeneous r_i remain permissible, but size-dependent change *within* a growing unit cannot generally be replaced by that unit's stock-level multiplier. Positive regrowth from zero stock is outside this representation, and intact-plot increments require endpoint calibration when r changes with state. Thus (19), not (22), governs the general application. Carbon units, atmospheric boundaries and overlap must be reconciled before either is used; a common CO₂ conversion can then be applied.

7.2 Regrowth from zero stock requires an increment basis

Secondary-forest recovery is a trajectory through age, composition and environment, not a fixed fraction of pre-clearing carbon. Neotropical observations document age-dependent and site-dependent biomass recovery after forest removal [21]. Accordingly, let cohort a occupy fixed area A_a , with baseline live-carbon densities c_{a0}, c_{a1} at two ages separated by $\Delta t > 0$. Let $c_{ak}^* = r_{ak} c_{ak}$ denote a specified endpoint recalibration. Areas in hectares and densities in MgC ha⁻¹ give gains in MgC yr⁻¹ when the interval is in years. For a selected non-decreasing baseline trajectory, define $g_a = (c_{a1} - c_{a0})/\Delta t \geq 0$. Its corrected increment satisfies

$$g_a^* = \frac{r_{a1}c_{a1} - r_{a0}c_{a0}}{\Delta t} = r_{a0}g_a + \frac{c_{a1}(r_{a1} - r_{a0})}{\Delta t}. \quad (23)$$

This is an age-cohort application of the endpoint identity, not evidence of temporal calibration drift. In particular, $c_{a0} = 0 < c_{a1}$ gives $g_a^* = r_{a1}g_a$: the cohort is retained and its initially undefined stock ratio need not be estimated.

Proposition 7.2 (Increment-weighted regrowth correction). *For the fixed cohort areas, let $I_+ = \{a : g_a > 0\}$, $I_0 = \{a : g_a = 0\}$, and $K = \sum_a A_a g_a > 0$. Define $q_a = g_a^*/g_a$ on I_+ and $v_a = A_a g_a^*/K$. Then*

$$K^* = \sum_{a \in I_+} A_a q_a g_a + K_0^*, \quad \frac{K^*}{K} = \mathbb{E}_v[q] + \frac{K_0^*}{K}, \quad K_0^* = \sum_{a \in I_0} A_a g_a^*. \quad (24)$$

Proof. Partition the sum $K^* = \sum_a A_a g_a^*$ into I_+ and I_0 , and substitute the definition of q_a on the former. The weights v_a sum to one because zero increments contribute nothing to K . $\square$

The averaging weights are *baseline gains*, not initial stocks. If $K_0^* = 0$, the replacement factor is a gain-weighted mean; a uniform endpoint correction supplies the familiar common-scale special case. A cohort with zero baseline increment but nonzero revised increment contributes additively and cannot be discarded. If $K = 0$, a total multiplier is undefined even when K^* is positive. A newly represented pool with zero baseline carbon at both endpoints also needs an additive model rather than a finite multiplicative correction. Baseline losses and corrected sign reversals require explicitly signed accounting instead of calling every term a removal.

For example, a zero-stock hectare that gains 5 MgC over one year gains 6 after a final-stock correction of 1.2; it remains part of the analysis despite having zero initial carbon. These are illustrative numbers. K is live-carbon accumulation over the selected interval, not gross photosynthesis or a complete atmospheric removal. Regrowth cohorts must be disjoint from any intact-forest accumulation subtotal, and mortality transfers, dead pools, soil exchange and exported carbon must be reconciled before K contributes to an atmospheric account.

7.3 Common-error cancellation and its limits

For a uniform stable multiplier r , fixed Z gives

$$\Delta N = (r - 1)(F - U), \quad \eta = \frac{|F - U|}{F + U}, \quad \frac{|\Delta N|}{|\Delta F| + |\Delta U|} = \eta \quad (r \neq 1). \quad (25)$$

η lies between zero and one. Thus a nearly balanced pair can undergo a large revision of both components with a small *absolute* adjustment of their net difference. The relative change of the nonzero calibrated net difference is still $r - 1$; cancellation must not be described as generally small relative net bias. Exact balance leaves the net unchanged, regardless of the common multiplier.

The converse sensitivity is just as important. Write $r_F = r + \delta/2$ and $r_U = r - \delta/2$. Then

$$N^* = r(F - U) + \frac{\delta}{2}(F + U) + Z^*. \quad (26)$$

A small differential correction is multiplied by the gross sum, not the net difference. Near balance, unequal source and sink corrections can dominate the common-scale contribution and reverse the sign of net exchange. Pairing therefore creates a testable mechanism for hidden gross bias, not an automatic means of making any proposed stock compatible with atmospheric constraints.

The rounded entries in [Table 4](#) provide a bookkeeping scale illustration: the positive labels sum to $3.7 + 2.2 + 1.1 = 7.0$, while the 5.5 removal magnitude and 0.6 closure remainder sum to 6.1, in $\text{GtCO}_2\text{e yr}^{-1}$. Their residual-to-absolute-sum ratio is $0.9/(7.0 + 6.1) \approx 0.069$, or roughly seven percent. The remainder absorbs omitted subcomponents and rounding, including the incomplete gas coverage of the organic-soil subtotal; these are neither independently extracted gross LULUCF fluxes nor ecosystem photosynthesis and respiration, and the illustration establishes no shared calibration error [12].

7.4 Size-dependent calibration and a directional census test

Size- and architecture-dependent discrepancies motivate a differential-correction test across all represented stem classes [1, 10, 22]. Large-tree studies provide one calibration domain, while the intermediate-stem contributions in Section 4.3 require their own event weights. Neither individual tree size nor a selected validation sample establishes the dominant contribution to population-level release, or that mortality receives a larger correction than growth. To see the distinction, let x be a tree's baseline carbon and let a stable, continuously differentiable, zero-intercept calibration be $m(x) = r(x)x$. A surviving tree growing from x_0 to $x_1 > x_0$ has gain multiplier

$$q_G(x_0, x_1) = \frac{m(x_1) - m(x_0)}{x_1 - x_0}, \quad m'(x) = r(x) + xr'(x). \quad (27)$$

The secant equals the average of m' over that interval. Mortality of a tree with assigned pre-death stock x_d , in contrast, contributes $m(x_d)/x_d = r(x_d)$. An increasing stock multiplier is not the same function as the growth multiplier. Species, architecture and changing organ allocation can require a multivariate calibration; the scalar construction isolates the logical distinction without imposing a universal fitted curve.

Let $G, L > 0$ denote positive gains and losses in a matched live-carbon census account, with net accumulation $S = G - L$. Sum calibrated survivor increments and consistently accounted recruitment into G^* ; use matched pre-death stocks and other specified live-pool losses for L^* . The resulting ratios $r_G = G^*/G$ and $r_L = L^*/L$ give

$$S^* = r_G G - r_L L = r_G S - (r_L - r_G)L, \quad S^* - S = (r_G - 1)G - (r_L - 1)L. \quad (28)$$

Recruitment across an inventory threshold is an entry into the observed pool, not necessarily carbon assimilated wholly during that interval. Diameter thresholds, survivor shrinkage, missing stems and death-time conventions must therefore be fixed or reconciled. Mortality here is loss from the live pool; atmospheric release follows its carbon-fate operator, not immediate emission by definition.

Directional hypothesis H_Δ . In a specified, independently calibrated census population, the event-weighted loss correction exceeds the gain correction by a preregistered meaningful margin, $r_L - r_G > \epsilon_\Delta \geq 0$. This predicts $S^* < r_G S$, lower net live-carbon accumulation than under common gain scaling. For S^* to be smaller than the *uncorrected* sink requires the stronger inequality $(r_L - 1)L > (r_G - 1)G$. Neither prediction follows solely from larger average size among dead trees. The hypothesis concerns measured event-weighted corrections. A prespecified interval for their difference wholly above the margin supports the directional claim; an interval wholly at or below the margin opposes it, while overlap leaves the test unresolved at that precision.

Table 7: Synthetic differential-census cases with $G = 120$, $L = 100$ and $S = 20$ in arbitrary carbon units over a common interval. The second row uses one increasing size-calibration function, not fitted forest observations.

Case	r_G	r_L	G^*	L^*	S^*
Loss-weighted hypothesis	1.20	1.50	144.0	150.0	-6.0
Increasing-size counterexample	1.85	1.50	222.0	150.0	72.0

The counterexample in Table 7 uses $m(x) = x(1 + 0.005x)$ in the specified synthetic units: 12 survivors grow from 80 to 90, while one larger tree of stock 100 dies. Then $r_G = 1.85 > r_L = 1.50$ and $S^* = 72$, despite the strictly increasing $r(x)$ and the larger dead tree. Underestimation of large stock levels therefore does not determine the corrected sink's direction. This counterexample does not deny differential amplification; it identifies the finite-increment evidence needed to turn that mechanism into a defensible prediction.

7.5 Operational census screens

For a common calibration applied to a matched census, let $\bar{x}_L$ be the mortality-stock-weighted mean tree carbon and $\bar{x}_G$ the gain-weighted mean of survivor interval midpoints. The quadratic family $m(x) = \alpha x + kx^2$, with $k > 0$, corrects losses more than gains exactly when $\bar{x}_L > 2\bar{x}_G$. A specified positive effect margin additionally requires the fitted magnitude of k .

For $m(x) = \alpha x^b$, $b > 1$, the exact finite-interval screen instead uses the ratio of loss- and interval-gain-weighted $(b - 1)$ -power means, which must exceed $b^{1/(b-1)}$. An arithmetic ratio below two excludes the quadratic direction but does not exclude every power exponent. Near $b = 1$, the log-moment difference determines the first-order direction and the differential tends to zero. Supplementary Note S2 gives the definitions, proofs, threshold behaviour and counterexample. These screens use baseline event distributions to identify admissible directions; independent calibration and uncertainty remain necessary to estimate their size.

7.6 Gross-source and net-land shares are different quantities

For a simplified account with other positive sources O and $Z = 0$, define

$$s_F^{\text{gross}} = \frac{F}{O + F}, \quad s_N^{\text{net}} = \frac{F - U}{O + F - U}. \quad (29)$$

The first is the forest fraction of the selected positive source sum. The second is a signed net-land contribution relative to a net total; it can be negative and becomes unstable as that total approaches zero. $F/(O + F - U)$ is a third ratio: a gross component divided by a net total. It is not a conventional composition of non-negative shares and can exceed one. When Z contains additional sources and removals, those components must be explicitly partitioned before forming a gross-source sum.

Table 8: Synthetic paired-correction example: $F = 6$, $U = 5.8$, $O = 50$, $Z = 0$. Flow units are arbitrary common annual mass units, not 2025 observations. The last two columns follow (29); percentages have different denominators.

Case	r_F	r_U	F^*	U^*	$F^* - U^*$	Gross share (%)	Net share (%)
Reference	1.0	1.0	6.00	5.80	0.20	10.71	0.40
Source only	2.5	1.0	15.00	5.80	9.20	23.08	15.54
Paired uniform	2.5	2.5	15.00	14.50	0.50	23.08	0.99
Paired differential	2.5	2.6	15.00	15.08	-0.08	23.08	-0.16

In the uniform example, corrections of +9 and +8.7 in gross emission and removal magnitudes change the net by only +0.3. The cancellation ratio is about 0.01695. With a slightly larger sink multiplier, the revised net becomes negative. These are algebraic demonstrations of identifiability and denominator dependence, not inferred global forest fluxes. They motivate independent constraints on source and removal weights, rather than a preferred multiplier.

7.7 Independent measurements remain decisive

A common allometric calibration can affect census-derived quantities, but it need not affect an independently measured atmospheric concentration increment, fossil-fuel statistic or tracer constraint in the same way. The theoretical cancellation concerns shared calibration of specified components; regrowth and live-census accumulation enter only after their increment operators and pool boundaries have been aligned. It cannot be imposed on independently constrained terms merely to force closure.

Likewise, living-biomass uptake, managed-land net removals and bookkeeping net land-use emissions are not interchangeable sink variables. A correction must follow the underlying mass estimator into its

valid pool and process, account for time lags, and identify overlap with other entries. Only then can the proposed source–sink revision be tested against a carbon-mass budget. The fact that a budget can contain compensating errors supplies no evidence for a particular compensation [13]. Without a supported offset, an upward revision of actual net forest release increases the corresponding reconstructed atmospheric accumulation, as made explicit in Section 8.1.

8 Atmospheric consequences and source attribution

8.1 Omitted emissions and atmospheric accumulation

A carbon-account correction can reveal a physically larger source than an earlier estimate represented. Let $M_{\text{atm}}(t)$ be atmospheric carbon mass, E_{DD} the actual CO₂–carbon release from the defined deforestation and degradation processes, E_{other} other sources, and $U_{\text{land}}, U_{\text{ocean}}$ positive removal magnitudes on a harmonized boundary. Conservation gives

$$\frac{dM_{\text{atm}}}{dt} = E_{\text{DD}} + E_{\text{other}} - U_{\text{land}} - U_{\text{ocean}}.$$

For two accounts or trajectories with the same initial atmospheric mass, define δ as corrected minus reference. Then

$$\delta M_{\text{atm}}(t_1) = \int_{t_0}^{t_1} [\delta E_{\text{DD}} + \delta E_{\text{other}} - \delta U_{\text{land}} - \delta U_{\text{ocean}}] dt.$$

A positive cumulative omitted forest release therefore gives a positive atmospheric-mass difference when the other trajectories are fixed. More generally, the difference remains positive whenever the cumulative source increment exceeds the cumulative removal increment. A reduction of uptake relative to a specified undisturbed baseline contributes through $\delta U_{\text{land}} < 0$ and can reinforce the release effect. That counterfactual loss of uptake must not be added again if already included in a net land-use term. The familiar source–sink partition in contemporary carbon budgets has this conservation structure [13].

Delayed decomposition changes the temporal profile of E_{DD} through the kernels in Section 6. Carbon released after the clearing year still enters the corresponding later atmospheric balance; products, charcoal and exports retain their specified fates. Additional uptake is a quantity to estimate, not a compensating adjustment to impose by definition. Consequently, the distinction between carbon stock and annual release does not remove the atmospheric consequence of genuinely omitted net emissions.

An emissions-based reconstruction and an observation-based atmospheric record are different estimators. Atmospheric CO₂ abundance is measured as dry-air mole fraction against reference gases; global burden and growth estimates additionally require spatial sampling, mixing and conversion assumptions [13, 23]. If released forest carbon was absent from a bottom-up inventory, it can nevertheless already be represented in the observed air. In that case, correcting the inventory explains a larger forest contribution to the observed accumulation and changes the source–sink reconciliation, rather than independently establishing an error of equal magnitude in the concentration record. Conversely, an emissions-driven projection that omitted that release can underestimate atmospheric accumulation. Observational coverage and calibration remain separately testable. These distinctions identify which estimate is underestimated without treating an inventory correction as physically inconsequential.

8.2 Underestimated totals and a potentially leading forest-loss contribution

Let F include the selected deforestation and degradation releases, with overlapping processes and carbon pools counted once. For a harmonized signed account $T = F + O - U$, with other signed terms absorbed into the defined O , the governing adjustment is

$$\Delta T = \Delta F + \Delta O - \Delta U. \quad (30)$$

Alternative I: an underestimated total. If $\Delta F > 0$ and $\Delta O = \Delta U = 0$, the reference account underestimates the total by ΔF . For a carbon-mass balance integrated over time, this also increases reconstructed atmospheric accumulation as shown in [Section 8.1](#). Unchanged sectors retain their absolute emissions while their fractions of the larger total fall.

Alternative II: a larger forest-loss attribution within a retained total. If $\Delta T = \Delta U = 0$, then $\Delta O = -\Delta F$. Forest loss receives a larger fraction, and the aggregate assigned to the remaining sources decreases. Sufficiently large, independently supported revisions can make deforestation and degradation the leading source in the selected account. The redistribution among other sectors requires their own observations; it is not fixed by the aggregate identity.

Alternative III: jointly reassess components. Sources, removals and total may change together. The paired correction in [Proposition 7.1](#) is a structured instance, not unrestricted uncertainty. Independent evidence can accept, constrain or reject it. None of the alternatives is selected solely by the shape of a ribbon.

For a comparison of positive source categories with other components O_j^* , the precise condition for forest loss to be the largest source is $F^* > \max_j O_j^*$. A majority of the selected positive source total requires the stronger condition $F^* > \sum_j O_j^*$. These are legitimate hypotheses under either Alternative I or II; a positive forest correction alone does not determine whether either threshold is crossed. Holding atmospheric accumulation fixed is also weaker than holding total emissions fixed unless removals are independently constrained. The domain, time window, treatment of removals and gas-equivalence basis must therefore accompany any source ranking. Ranking annual GHG emissions does not, by itself, allocate cumulative historical warming among causes.

9 A calibration-first programme for forest-carbon accounting

9.1 Matched measurements and explicit residuals

A discriminating study compares both estimation chains with a common, independently obtained material reference. Record species, individual geometry, disjoint organ volumes, basic-density samples and tissue-carbon fractions across the relevant size and disturbance classes. Preserve noncommercial material and separately identify dead pools. On common plots and dates, let C_{ref} be that independently checked reference, and define residuals $e_A = \widehat{C}_A - C_{\text{ref}}$ and $e_R = \widehat{C}_R - C_{\text{ref}}$ for the allometric and reconstruction pathways. Report signed errors, magnitude, uncertainty and residual dependence on crown architecture, organ allocation and disturbance, rather than only correlation.

A negative allometric residual associated with omitted architectural characteristics supports the proposed underestimation mechanism; an independent reconstruction with negative or positive residuals requires its own correction. Joint uncertainty includes errors shared by volume, density, chemistry and map registration. A prespecified equivalence margin can distinguish agreement from scientifically meaningful overestimation or underestimation, while overlapping intervals remain unresolved. No estimator is granted accuracy merely because it supplies the calibration labels of another.

9.2 Calibration and spatial extension

Build vegetation-specific reference distributions before extrapolating across biomes. Withhold complete sites or vegetation strata before any label construction or parameter tuning. Retain the acquisition period of field observations and predictors, the spatial support of each label, and the geographic extent over which interpolation or extrapolation occurs. Evaluate predictor resolution and saturation against independently reconstructed material, rather than interpreting predictor saturation as proof of one error sign. Outside the calibration support, mark uncertainty or withhold prediction instead of assuming that additional pixels establish material accuracy.

The original pantropical result supplies a worked spatial application of this chain. Future refinement can improve material labels, their biological coverage or their spatial transfer. A revised calibration outcome would update the corresponding estimate, while the conditional identities linking labels, carbon and release remain valid on their stated domains. Supplemental class-density and volume sensitivities provide reproducible comparison diagnostics; they are not substitutes for representative records or an argument that the original estimate has been empirically rejected.

9.3 Disturbance-weighted validation and annual observations

Carbon lost through deforestation and degradation must be sampled where those processes operate. Include intermediate stems, large trees, partial crown losses, collateral mortality and repeated damage, with affected fractions applied to remaining material only once. Assign fuel, products, dead wood and other fates explicitly, and estimate regeneration and changed uptake on separate, consistent boundaries. At matched supports, evaluate the covariance between carbon-calibration residuals and disturbance or uptake weights. It is that association, not an unweighted map-average discrepancy, that transfers the correction into the annual source population.

Apply repeated-census calibration to both endpoints of surviving trees and to the specified stocks of mortality events. Recruitment across a measurement threshold needs its own observed-pool convention. The operational screens in Section 7.5 identify admissible directions within a chosen calibration family; independent reference data are still needed for the size of the correction. Regional atmospheric observations and independent source/sink information constrain the resulting physical account [24]. They should not be used to introduce an unmeasured compensating sink simply to close an assumed total.

9.4 From physical exchange to inventory attribution

A carbon stock, a gross atmospheric release and a GWP-weighted annual inventory are different quantities. First reconstruct physical source and removal histories; then map those histories into the definitions of the chosen reporting system. EDGAR's signed land categories and the Global Carbon Budget's land-use flux use different attribution rules. Grassi et al. demonstrate that harmonizing environmentally induced sinks on managed forests can strongly change an aggregate comparison [14]. Such harmonization does not determine true tree mass. Conversely, revising tree mass does not by itself select an inventory convention.

A carbon-conservation constraint can be written as

$$G_{\text{atm}} = E_{\text{fossil}} + F - U - S_{\text{ocean}} + R, \quad (31)$$

where all terms use the same carbon units, period and boundary. With other terms fixed, this constrains $F - U$ rather than identifying F and U separately. The change $(F, U) \mapsto (F + a, U + a)$ preserves that one balance where admissible. Additional regional, temporal and tracer observations can constrain this ambiguity. Neither a closed global balance nor a precise remote map establishes an independently correct partition of the underlying components.

10 Discussion and conclusions

The primary issue is the carbon scale inherited from the measurement chain. Allometric relationships can be informative predictors, yet a transferred, reduced-input relationship can misrepresent material architecture and composition in a target forest. Direct organ-level reconstruction addresses that mechanism by measuring quantities hidden inside a generalized mass prediction. The architectural and tissue-based method underlying the published pantropical reassessment provides an application of this approach, supported by its reported physical and laboratory tests at their documented scales.

The LiDAR distinction follows from the same logic. Greater spatial coverage and accurate canopy ranging do not by themselves remove error in biomass labels. Equation (3) identifies the component

transmitted by an ideal label-trained predictor; equation (4) additionally separates transfer between populations. Validation against independently measured material is consequently different from cross-validation against labels derived from one biomass equation. Detailed laser reconstruction is a constructive route to such material information, not an exception to be ignored. Sensor coverage and calibration quality must be evaluated together.

Calibration validity is population- and pathway-specific. A relationship supported on one forest architecture need not transfer unchanged to another; a TLS-derived material reference has different dependencies from a waveform regression fitted to biomass labels. Validation therefore targets the measured carbon scale and its transport. The methodological failure arises when allometry-derived labels are treated as independent physical truth and agreement with those labels or increased spatial coverage substitutes for an independent material test. Bias must be investigated at its measurement source, alongside improvements in spatial precision.

The material boundary is equally consequential. Carbon in nonmerchantable branches, tops, buttresses, smaller stems and damaged material remains part of the designated biological pool. A timber-volume convention is not a total-carbon measurement. At the same time, many biomass equations intend to represent total material through empirical calibration; their failures must be established by measured residuals and population support rather than a misdescription of their target. Architecture and organ allocation are mechanisms of allometric error, not a set of competing explanations that displace it.

The 723.97 PgC estimate is retained as the published aboveground-carbon application for its mapped pantropical domain, excluding the Australian tropics, at a nominal 2007–2008 map epoch. This article does not replace that total with a conventional inventory or produce a new world stock. It distinguishes reported measurement evidence, spatial calibration, source-table provenance and the additional observations needed for an annual account. Calibration can continue and improve the estimate without making one immutable number a prerequisite for the measurement thesis. The supplemental table correction concerns class labels and leaves reported stock amounts unchanged.

Annual consequences depend on the carbon actually affected and its fate. The stock-to-flux identity exposes disturbance-weighted calibration bias; cohort convolution preserves decomposition and product lags; paired accounting distinguishes source and removal corrections rather than presuming common cancellation. All stem classes and degradation pathways contribute through abundance, affected material and release histories. An omitted positive net source increases reconstructed atmospheric accumulation unless other changes offset it. If the total and removals are retained, an upward forest-source correction instead increases the attributed contribution of forest loss. Whether that contribution is largest or a majority remains a quantitative question under the specified account.

The resulting research programme begins with independently checked material, extends calibrated labels through declared biological and spatial domains, and propagates the resulting carbon into explicit source–sink operators. The methodological advance is this connected chain of measurement and inference: improved coverage becomes improved carbon accounting only when the physical calibration and its transfer are made equally explicit.

Data and code availability

The companion package contains the complete article, technical supplement, transcribed source inputs and executable numerical analyses. The publication distribution comprises the compiled documents, three figure plates and computational support identified in `PUBLICATION_MANIFEST.json`; editable document sources are supplied separately. The project repository is https://github.com/hyrucaji77/Tropical_Forest_Carbon-; edition and publication metadata accompany the distributed files. Numerical verification reproduces the stated identities and source transcriptions. Original-record limitations are

described in SOURCE_AVAILABILITY.md. No archival DOI has been assigned.

Author contribution

Henry Arellano-Peña is the sole author and is responsible for the analysis, interpretation and presentation.

Competing interests

Henry Arellano-Peña owns the methodologies discussed and is founder and director of NEBIOT S.A.S., the sole company affiliated with this work. NEB001 provides the vegetation-mapping framework and NEB002 the architecture, material-analysis and calibration chain underlying the published reassessment [1, 9]. These relationships are disclosed; the author declares no other competing interests.

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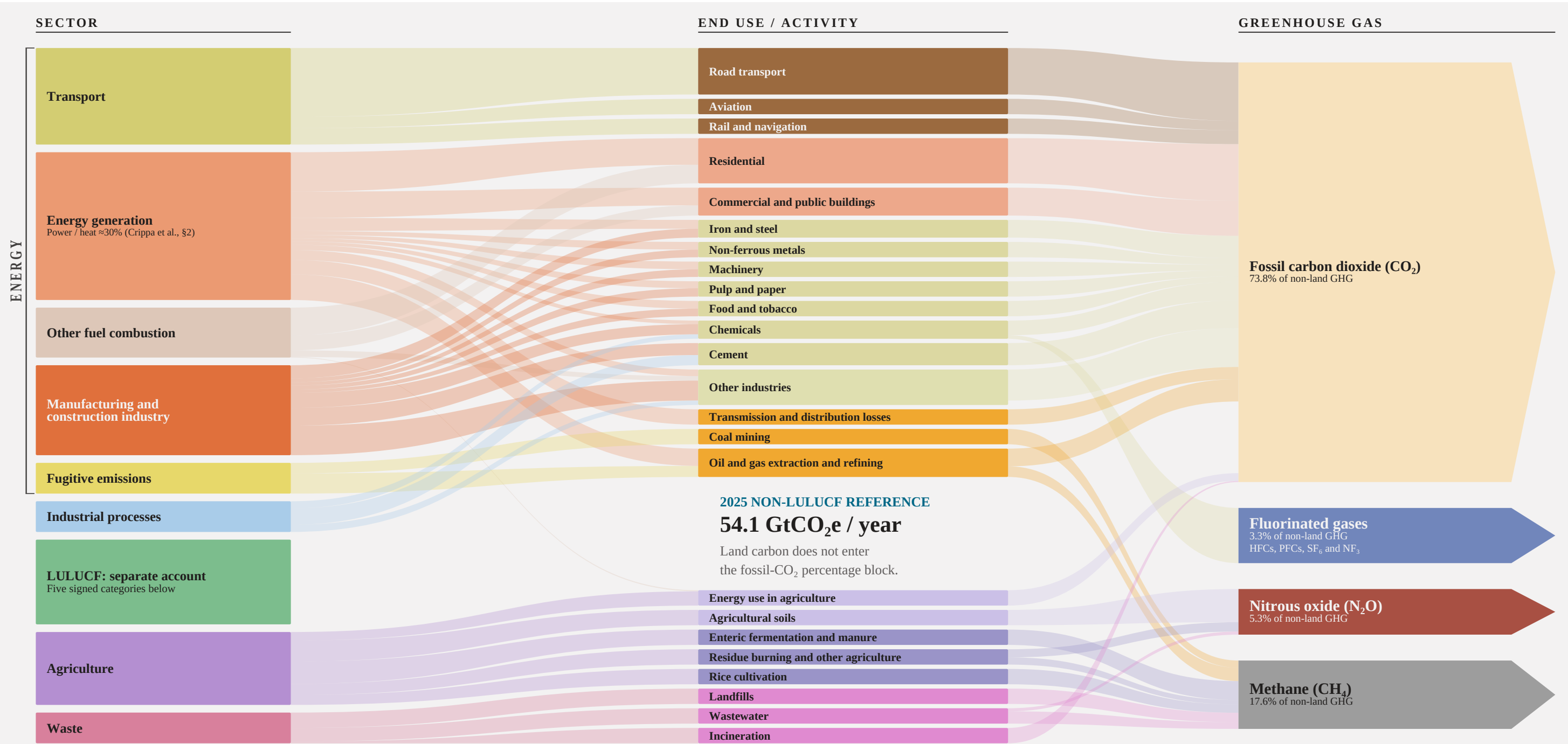
Use of generative artificial intelligence

ChatGPT (OpenAI) assisted with drafting, mathematical exploration, source discovery, code and LaTeX preparation. Claude (Anthropic) generated critical-review material used during manuscript development. Model outputs were assessed against the identified sources and analytical derivations rather than treated as empirical evidence. Scientific responsibility remains with the named author.

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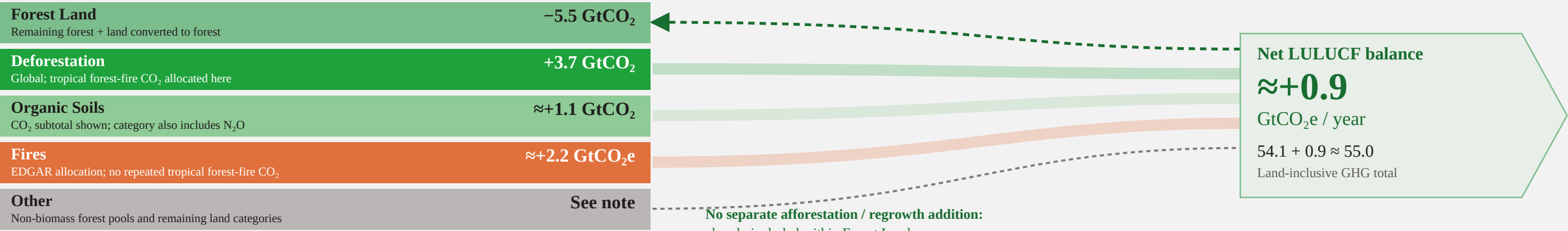
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THE SIGNED LAND ACCOUNT

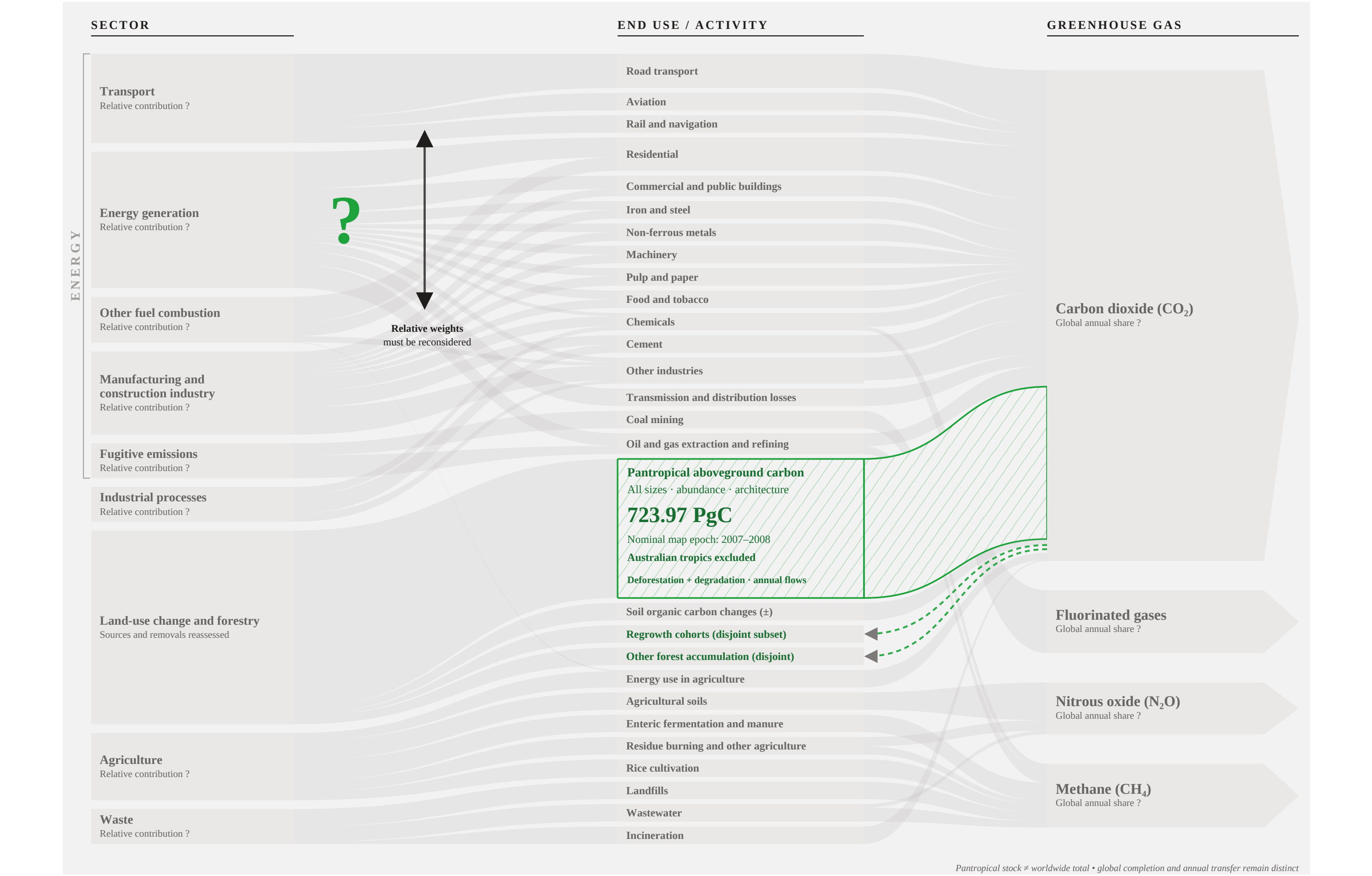
EDGAR 2026 · Section 3, Figure 6 and Annex 2 · 2025 estimates



Other / rounding remainder: ≈-0.6 GtCO₂e is a closure diagnostic, not an independently extracted category total.

Disjoint land categories, separate gas denominator. Forest Land already includes land converted to forest; no extra afforestation or regrowth subtotal is added. Deforestation includes tropical forest-fire CO₂, which is excluded from Fires; tropical fire non-CO₂ gases remain in Fires. Organic Soils’ quoted 1.1 is CO₂ only. The diagnostic remainder 0.9 − [3.7 − 5.5 + 2.2 + 1.1] ≈ −0.6 also absorbs omitted subcomponents and rounding; it is not a separately extracted Other estimate. Source: Crippa et al. (2026), booklet Sections 2–3 and Annex 2, doi:10.2760/7717504 (CC BY 4.0).

Figure 1: The reported 2025 account and its boundaries. The sector–activity–gas organization is adapted from WRI (2005) and Ecofys (2013), with author-created vector artwork and a separate five-category signed land account. Intermediate activity nodes describe pathways, rather than EDGAR’s official sector list. Gas shares refer only to EDGAR’s 54.1 GtCO₂e non-LULUCF inventory using AR5 GWP₁₀₀; 73.8% is fossil CO₂. Land emissions therefore do not enter that gas block. The reverse dashed link denotes Forest Land’s net removal. Forest/deforestation estimates use a global Tier 1 approach with default parameters and estimated activity data beyond the latest input maps, not independently remeasured 2025 carbon densities. Paths and all graphical areas are schematic, not a balanced quantitative Sankey. Electricity-attribution paths must not be counted again as direct emissions.



All sizes, not only the largest trees. In the *Acalypha–Guazuma* community, the 10–30 cm class contains 38.97% of reported biomass in a community with total 147.8 Mg ha^{−1}; 2.88 t divided by 30 stems gives a derived mean of **96 kg biomass per stem** (Vásquez and Arellano, 2012, Table 170, p. 942). This older class distribution is allometry-based and illustrates abundance rather than class-5 density; the separate architectural records near 100 kg illustrate that method’s size coverage. Abundance, affected fractions and release histories govern population contribution.

Physical consequence and attribution. A positive net source missing from an account leaves actual accumulation above the account’s reconstruction unless other terms offset it. With total and removals held fixed, forest loss receives a larger share and can become the leading source if its corrected flux exceeds competing categories. Delayed release enters later years. Observed air can already contain emissions missed by an inventory; changing attribution does not itself establish an equally large measurement error in atmospheric concentration.

SPATIAL COVERAGE: THE PANTROPICAL RESULT IS NOT A WORLDWIDE TOTAL		
Mapped contribution: 723.97 PgC Aboveground carbon; Australian tropics excluded.	Additional regional stock Australian tropics and extratropical vegetation require separate estimates.	Worldwide completion Add compatible pools and dates; annual net exchange remains separate.
Nominal output-map reference: 2007–2008 (preprint p. 23). Baccini’s 2012 publication describes a 2007–2008 map; Simard’s height predictor uses 2005 GLAS data. Forest-only totals require a common forest mask.		
HISTORICAL 2010 ILLUSTRATION — NOT A 2025 EMISSIONS ESTIMATE		
Fixed total: 26.16–32.61%. Other contributions retained: 22.58–26.66%. Both impose the stock multiplier on Ecofys’s combined land-use category. The published 26.15–32.60% endpoints are reproduced by truncation; alternative hidden precision is possible. Supplementary Note S3 gives the source, assumptions and calculation.		

Figure 2: Pantropical architectural measurement and worldwide annual attribution. The green branch identifies 723.97 PgC as the mapped pantropical aboveground-carbon estimate, explicitly excluding the Australian tropics (Arellano-P. and Rangel-Ch., 2016, pp. 23–24). Its 2007–2008 nominal map epoch differs from publication and predictor-acquisition dates (Baccini et al., 2012; Simard et al., 2011). Global coverage additionally requires omitted tropical and extratropical vegetation on matched pool, mask and date definitions. Hatching denotes unresolved annual transfer; the label retains the published standing stock, and graphical areas are schematic. The organization adapts the WRI/Ecofys source–activity–gas convention. Regrowth and other accumulation are disjoint conceptual subsets, not extra EDGAR subtotals. Muted sectors, the question mark and opposing arrows express a larger source total and reconstructed atmospheric burden under fixed other terms, a larger forest-loss attribution within a retained account, or joint reassessment. All stem sizes and partial degradation are included; no source ranking is inferred from graphical width. The historical inset is conditional 2010 arithmetic, not a measured 2025 allocation. Figure 3 separates allometry-calibrated spatial prediction from architecture-informed material measurement. The article distinguishes measurement evidence from annual-transfer conditions; source-table and class-level comparison details are in the technical supplement.

CARBON ACCURACY STARTS WITH THE REFERENCE

Spatial coverage measures where; a validated material conversion determines how much.

A • PROXY-LABEL PATHWAY

Reduced predictor measurements

Diameter, canopy height, nominal density and context.
Organ architecture may remain only implicitly represented.



Allometric biomass / carbon labels

A fitted response summarizes its calibration population.
Transport can fail when architecture or composition changes.



Remote-sensing calibration

LiDAR or optical metrics are fitted to reference labels.
Accurate prediction of those labels is not independent mass validation.



Expanded spatial prediction

More coverage can reduce sampling or mapping uncertainty.
A shared conditional carbon-scale bias can remain.

INHERITANCE IDENTITY • IDEAL CONDITIONAL PREDICTION

$E[\text{label} \mid \text{LiDAR}] - E[\text{true carbon} \mid \text{LiDAR}] = E[\text{reference-label bias} \mid \text{LiDAR}]$

Conditional mean-zero residual noise is assumed. This is a mechanism, not an empirical global correction factor.

Published methodological application

723.97 PgC · mapped pantropical aboveground carbon
Nominal map epoch 2007–2008; Australian tropics excluded.
221 reconstructed trees; 2,232 tissue samples reported.
Physical-volume checks and spatial calibration are distinct steps.

B • ARCHITECTURE-INFORMED PATHWAY

Measured organs and material

Stems, branches, bark and other explicit compartments.
All stem sizes; biological volume, not saleable timber only.



Carbon from geometry and tissue

Carbon = Σ (organ volume × basic density × carbon fraction).
Direct laser reconstruction may help measure woody volume.



Independent calibration and holdouts

Validate material geometry, density, chemistry and sampling.
Keep reference sites separate from model fitting.



Spatial extension with traceable targets

LiDAR can extend a separately validated carbon reference.
Check transfer, omitted structures and regional extrapolation.

From carbon measurement to annual accounts

All sizes × abundance × affected fractions × release histories.
Deforestation, degradation, regrowth and other removals.
Uncounted net release raises reconstructed accumulation.
A retained total constrains reattribution among sources.

NOT EVERY LiDAR PRODUCT USES THE SAME MEASUREMENT CHAIN

Proxy-based biomass mapping and direct terrestrial 3D material reconstruction are different pathways.

Measure architecture and organ allocation to identify the source of allometric calibration error.

Refining calibration can improve the stock estimate without changing the distinction between reference accuracy and coverage.

Sources: Arellano-P. & Rangel-Ch. (2016); Goodman et al. (2014); Calders et al. (2015); GEDI L4A V2 documentation.

Operational diagram: no graphical width represents a measured carbon share. Henry Arellano-Peña.

The calibration distinction. A remote-sensing product fitted to allometric reference labels can inherit their conditional bias. Direct reconstruction of biological material with measured tissue properties provides a different reference pathway, which can itself use terrestrial laser measurements. Independent reference tests are required for both approaches; greater coverage alone does not determine absolute carbon accuracy.

Figure 3: Two measurement-to-map pathways and their annual-accounting consequences. The left pathway extends a calibrated proxy; the right begins with organ geometry and matched material properties. Architecture and tissue variation identify mechanisms by which an allometric predictor can depart from actual biological carbon. The conditional-expectation identity assumes a source distribution and mean-zero residual noise conditional on the sensor variables; a real learner also has fitting and transfer errors. The original reconstruction and physical tests are documented by Arellano-P. and Rangel-Ch. (2016). Crown-related biomass discrepancies are independently illustrated by Goodman et al. (2014). Calders et al. (2015) demonstrates a terrestrial laser material-volume route, while the GEDI L4A Version 2 guide documents a model linking waveform height metrics and biomass reference data. These examples retain their study domains and establish no universal multiplier. Total material volume includes nonmerchantable woody organs; belowground and dead pools remain separately defined. The reported 723.97 PgC is a geographically and temporally delimited standing stock. Its annual implications require affected populations, release histories and removals. Neither larger spatial coverage nor numerical reproduction of calibration labels is equivalent to independent carbon-mass validation. Original diagram: Henry Arellano-Peña.